

FINANCIAL DEVELOPMENT AND ECONOMIC GROWTH: EMPIRICAL EVIDENCE FROM ECCAS¹ COUNTRIES

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Abstract: The target of this paper is to empirically assess the influence of financial development on the economic growth of the ECCAS zone for the period 1990-2012. For that, we followed the approach of Berthelemy and Varoudakis (1998) which consisted in modelling a log-linear dynamic panel equation. So, the Pedroni cointegration test showed that there is a long term equilibrium relationship between growth and the financial system in that zone. In others words, the standard of living improves as a consequence of the financial deepening. Also, estimations through the System Generalized Methods of Moments (System GMM), proved that: (i) financial intermediation positively influences the economic growth but not in a significant way; (ii) political instability is detrimental to the well-being of individuals. In line with these results the following recommendations are formulated: to encourage the pursuit of financial reforms and the development of financial systems in that zone; to finance innovation, technical and vocational training in order to diversify production structures and exports; and to improve the business environment which is a prerequisite for the dynamism of the private sector.

Key Words: Financial Development, Economic Growth, System GMM, ECCAS.

JEL Classification : C23; G28; O16; O40.

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¹ The Economic Community of Central African States (ECCAS) is one of the Regional Economic Communities in Africa. It is composed of ten countries namely: Angola, Burundi, Cameroon, Central African Republic, the Democratic Republic of Congo, Congo, Gabon, Equatorial Guinea, Sao Tome and Principe, and Chad.

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1. INTRODUCTION

Over the last two decades which ended, Africa has experienced a problematic economic and financial situation. Indeed, the relative increase in growth rate has not produced the expected results in terms of improving the wellness and the living conditions of populations. Rather, there has been an increase in debt, government deficits, impoverishment and inequality.

In 2000, the adoption of the "Millennium Development Goals" (MDGs) highlighted the priority of development issues. That is why several African countries have developed plans for economic development by incorporating all the priority areas of the MDGs. But those plans, which serve as a compass for the development and the orientation of public policies, have neglected the role of the financial sector.

Indeed, Gurley and Shaw (1955) think that the financial aspect of the development is sometimes overlooked. However, the literature on the relationship between finance and growth is very rich and controversial. It has established the different mechanisms through which the financial system influences economic growth on both the theoretical and practical aspect. According Beitone and *al.* (2010), financial intermediation is to provide payment services, mobilize savings and allocate credit. Meanwhile, economic growth is a complex concept which represents "an increase in a dimension indicator, GDP, for example"; in the view of François Perroux (1961).

Many authors have outlined the importance of finance in the economic growth process. Amongst them, we have Bagehot (1873) and Schumpeter (1912). According to the first author, the rapid development of the Great Britain was due to the superiority of its financial market in the mobilization of savings towards the financing of long-term investments. For the second author, bank credits are essential to economic growth and innovation; and the banker needs to fund entrepreneurs who have the best chance to succeed in their projects.

But, authors such as Robinson (1952), Lucas (1988) and Levine (2003) do not agree with the conclusions above. According to them, finance has no effect on economic growth. In fact, for the neoclassical theory, money cannot affect real activity. Some proponents of this approach think that finance can follow economic growth (Lewis, 1995; Levine and Zervos, 1998); or influence one another at the same time (Cameron, 1972).

It is with the endogenous growth theories that the relationship between finance and growth obtained theoretical design. Therefore, the development of financial systems affects economic growth by acting on three factors namely: the capital productivity, the efficiency of financial systems and the savings rate. Pioneers models are those of Bencivenga and Smith (1991); joined by Pagano (1993); and Berthélemy and Varoudakis (1994).

Since the theory without practice is useless, some authors conducted empirical verifications about that relationship. Firstly, King and Levine (1993) innovated in experimental studies by using, for the first time, panel data in their paper. They showed that better financial systems stimulate both economic growth and per capita productivity by efficiently allocating capital. Secondly, the analysis of convergence clubs made by Berthélemy and Varoudakis (1996) has shown that financial development affects economic growth if and only if the country has a M2 to GDP ratio greater or equal to 36.5%. Thirdly, Beck and *al.* (2000) showed that the exogenous development of banks is positively correlated with economic growth in their sample. They empirically evaluated the dynamic relationship between the level of financial development, the economic growth, and a set of explanatory variables. Fourthly, concerning Africa, Kpodar (2004) showed that financial development has a lower and marginal impact on economic growth in Sub-Saharan Africa because of the following reasons: the level of ethno-linguistic fragmentation, the recurrence of banking crises, the banking concentration, and the government intervention.

Given the explanations above, and after decades, one might question the validity of these results between finance and growth especially in the context of sub-Saharan Africa. In fact, the majority of African studies have neglected the importance of regional economic communities. Those

entities play a decisive role in the implementation of development policies adopted by the African Union. Therefore, we agree with Bekolo-Ebe (2002) to say that the banking intermediation function appears as a fundamental element of economic growth and development process. So, our paper attempts to answer the main question: *What is the influence of financial development on economic growth of ECCAS countries?*

Hence, the objective of this paper is to contribute to the debate on the relationship between financial development and economic growth. Our study is an empirical one and targets variables at the macroeconomic level. Studies on ECCAS exist namely those of Avom and Carmignani (2008b; 2010), but there is no study on the finance and growth nexus. Our work helps to fill that void. Our main hypothesis follows that of Levine (1997). In fact, for Levine (1997), financial development contributes positively to economic development.

Our paper presents multiple interests. Firstly, ECCAS is one of the richest areas in Africa in terms of natural resource endowment. However, access to and cost of financing does not facilitate investment in these countries according to the World Bank (2008). Secondly, ECCAS has the vision to emerge in 2025. It is therefore important to see to what extent the financial system can contribute to this process in order to avoid some negative effects like banking and financial crises.

In terms of methodology, and to analyze the effects of financial deepening on the level of GDP per head of ECCAS countries, we followed the dynamic panel modelling made by Berthélemy Varoudakis (1998). Financial variables used in our work are essentially those of the banking sector in the sense that the other sectors (microfinance, insurance, tontine, etc.) do not have adequate, reliable and available data. Similarly, the data collected ranges from 1990-2012. This is to assess the contribution of the financial sector reforms started in 1990. These data were obtained from the *World Development Indicators* (2013). However, estimates have been conducted only on nine countries in order to work on a balanced panel. Also, the coefficients were estimated by using the generalized method of moment's technique (*GMM System*) which helps to avoid spurious regression.

Overall, our results show that financial development positively affects economic growth in ECCAS. But this influence is not significant. This result is in line with that of Avom and Carmignani (2010). Also, we find that political instability is detrimental to the process of economic growth in the area. This result corroborates the findings of the joint report RAIAL, Oxfam, and Saferworld (2007) on the harmful effect of conflicts on the living standards in the region.

The rest of the article is organized as follows: the second section provides a brief literature review on the finance and economic growth nexus. In the third section, we present the methodology and some stylized facts. The section four highlights and discusses the results while the section five concludes.

2. LITERATURE REVIEW

The various schools of thought in economics do not agree on the role of financial intermediaries in the process of economic growth. Bagehot (1873) and Schumpeter (1911) support the idea that finance is a necessary condition for economic growth. But Levine (2003) states that finance has no effect on economic growth. Some authors even suggest that finance follows growth (Robinson, 1952) while others argue that the role of finance does not matter (Lucas, 1988). This debate is relevant on both the theoretical and empirical areas.

2.1. Finance and economic growth: the theoretical debate

Gurley and Shaw (1960) believe that the main function of financial intermediaries is to transfer the fraction of unexpended income, from surplus agents, to agents in need of financing. This

helps to improve the level of savings and investment; and a more efficient allocation of capital. Then, the diversification of financial instruments leads to an optimum economic growth.

The banking sector plays a crucial role in the economy according to Gerschenkron (1962). For this author, the underdeveloped economies need an active financial sector in order to realize an economic takeoff. Therefore, the importance of the banking system in the economy increases with the backwardness of the economy.

Following the work of their predecessors, Bencivenga and Smith (1991) constructed a model in which the behavior of financial intermediaries influences the allocation of resources in any given economy. Also, savings behaviors influence the performance of the growth rate. In their model, only economies with an active financial sector can experience greatest growth rates. This result has been validated by Pagano (1993) in his seminal contribution. Then, many authors have attempted to empirically verify those theories.

2.2. Finance and economic growth: a divergence in empirical results

Berthélemy and Varoudakis (1996) used panel data to provide a benchmark on the convergence clubs in their sample size. They showed a non-linearity between the financial development indicator (liquidity ratio) and the economic growth (measured by the Real GDP). In fact, beyond a certain threshold, countries move from one convergence club to another. And, it positively affects the growth rate. But, their panel data estimating used a bad specification.

That is why in a later study; Berthélemy and Varoudakis (1998) assessed the contribution of financial development to economic growth through financial reforms. Methodologically, they estimated a fixed effects model on an unbalanced panel made of 82 countries from 1960 to 1990. Their results did not validate the positive influence of finance on growth. To explain these results, the authors thought about the existence of threshold effects associated with multiple equilibria. Thus, they claimed that financial development has an impact on economic growth if and only if countries have a M2 to GDP ratio greater or equal to 36.5%.

In its work, Kpodar (2005) noticed that sub-Saharan Africa (SSA) is marginalized in empirical studies. By retracing some contributions of authors about Africa, he retained 24 African countries among the sample of 64 countries in his study. Thus, the estimation by the Generalized Method of Moments has shown that financial development has a lower and marginal impact on economic growth in SSA. To explain this phenomenon of marginalization, the author advanced the following reasons: ethno-linguistic fragmentation, banking crisis, banking concentration, and government intervention.

However, Khatkate and Bhatia (1975) did the first work on Africa. The analysis done on a sample of 11 countries for the period 1960-1970, showed that the correlation between finance and economic growth is positive in countries like Kenya, Zambia and Ivory Coast. Conversely, that correlation is negative (or absent) in countries like Ghana, Sierra Leone and Mauritius.

In the study of Spears (1992), the correlation between finance and growth is positive for the majority of countries in their sample except for Zambia and Malawi. His results are more conclusive than that of Khatkate and Bhatia (1975). He used graphs and correlation coefficients for 10 countries from 1966 to 1986. The financial development indicators consisted in: (i) ratio of demand deposits and savings accounts on M2; (ii) M2 and M3 aggregates as percentage of GDP. The conclusions of his work have joined those of Odedokun (1996), Gelbard and Leite (1999), Collier and Gunning (1999), Ndikumana (2000), Ghirmay (2004) and Meagon (2005).

Finally, the study done by Raffinot and Venet (1998) on the Western African Economic and Monetary Union (UEMOA) showed that results about finance and growth are not satisfactory. The results are not satisfactory when non-significant variables are used in the analysis. They include: Trade Openness, Log of GDP, Enrollment Rate and Rate of Investment. They found that the correlation between M2/GDP ratio and economic growth is 2%.

The divergence of the above empirical results shows many things. Firstly, there is no study on the relationship between finance and growth for ECCAS countries. Our paper wants to fill that void. Secondly, the majority of studies on Africa neglected the role played by Economic Regional Circles. In fact, they are in charge to implement the development policies adopted by the African Union. Thirdly, ECCAS wants to emerge by 2025 and it is important to assess the role of the financial system in achieving the vision 2025. All these reasons motivated our choice of ECCAS.

3. METHODOLOGY

The econometric of panel data seems to be a relevant analysis process. In fact, it includes both individual and temporal dimensions. This section of our paper discusses: the model, the variables and their expected signs, the sources of data and the estimation procedure.

3.1. Model

We follow the log-linear dynamic panel specification like made by Berthélemy and Varoudakis (1998). These authors followed the modelling of Mankiw, Romer and Weil (1992). But our specification differs from the first authors in terms of dummy variable. The final model that we estimate is written as follow:

$$GDPC_{i,t} - GDPC_{i,t-1} = \alpha_0 i + \beta_0 GDPC_{i,t-1} + \beta_1 LM2_{i,t} + \beta_2 LCREPRI_{i,t} + \delta_0 LINFL_{i,t} + \delta_1 LINV_{i,t} + \delta_2 LEDU_{i,t} + \delta_3 LTOP_{i,t} + \lambda CONFL_{i,t} + \varepsilon_{i,t}$$

All variables are expressed in logarithm except the dummy variable representing the conflict (CONFL). The dependent variable is the GDP per capita (GDPC). Variables M2, CREPRI, INV, TOP respectively represent: money supply in the sense of M2, credit to the private sector, the level of private investment (gross fixed capital formation) and trade openness (exports plus imports). These variables are expressed as a percentage of the GDP.

We use the two famous indicators of financial development namely M2 and CREPRI. We follow, with these indicators, the works of Berthélemy and Varoudakis (1998); Kpodar (2005). INFL refers to the level of inflation measured by the consumer price index (base 100 = 2005). Human capital, noted EDU, is approximated by the number of years in the high school.

The country index is given by i , while t represents the index of time. The index i vary from 1 to 9 and the index t varies from 1990 to 2012. The error term is represented by ε . The coefficient α is the constant term while the coefficient λ represents semi-elasticity. We analyze the coefficients β_i ($i = 0,1,2$) and δ_i ($i = 0,1,2,3$) as elasticities.

3.2. Variables and expected signs of the coefficients

We summarize all the different variables used in the analysis as well as signs of their expected coefficients in the table below. Descriptive statistics relating to the variables are in appendix 1 of this article.

Table 3.1: Variables and expected signs of the coefficients

Variables	Expected signs of the coefficients
M2 (Money Supply)	+
CREPRI (Bank Credit to the Private Sector)	+
EDU (Human Capital)	+
INV (Private Investment)	+
INFL (Inflation Level)	+ / -
TOP (Trade Openness)	+
CONFL (Political Instability, <i>dummy</i>)	-

Source: inspired from Beth  l  my and Varoudakis (1998)

3.3. Sources of data

The data come from the *World Development Indicators* (2013). The period of study ranges from 1990-2012. Data on conflicts come from the joint report RAIAL, Oxfam, and Saferworld (2007). The sample of selected countries is 9 countries (exception of Sao Tome and Principe). Indeed, data for that country are inadequate and non-existent for certain variables of our model. That is why we have excluded this country in the sample in order to work on a balanced panel.

3.4. Estimation procedure

To estimate the parameters of our model, we conducted several preliminary tests namely: unit root test in panel and cointegration test panel. The results are given below while the respective tables and statistics are given in the appendices.

3.4.1. Unit root tests in panel

We chose, in our work, to use the IPS test of Im, Pesaran and Shin (2003) in the sense that it is stable, efficient, and responsive to small panel data.

Test Hypotheses: H0: the series is not stationary; against H1: the series is stationary.

Decision Rule: We accept the assumption of non-stationarity of the series when the probability associated to the test is greater than 5%. However, when this probability is less or equal to 5% it is concluded that the series is stationary.

The results of the unit root test (see Table 3.2) show that trade openness (TOP), the level of investment (INV) and inflation (INFL) are integrated of order zero. The variables representing the GDP per capita (GDPC), money supply (M2), education (EDU) and credit to the private sector (CREPRI) are integrated of order one.

So, to avoid spurious regression between variables integrated of order zero and variables integrated of order one, we proceed to the cointegration test of Pedroni (1999). This test helps to check if there is a long-term equilibrium relationship between those variables.

3.4.2. Pedroni cointegration test in panel (1999)

This test considers both the notions of inter-individual cointegration and intra-individual cointegration. From the seven tests proposed by Pedroni (1999), four are based on the dimension *within* and three are based on the dimension *between*.

The two categories of tests (*within* and *between*) are based on the null hypothesis of no cointegration. This hypothesis is accepted if the value of "Prob" or "P-value" is greater than 5%. In the case where this value is less or equal to 5%, we accept the alternative hypothesis of the presence of cointegration.

The results of Pedroni cointegration tests (Table 3.3) show that there is a long-term equilibrium relationship between our variables. Indeed, from the seven test statistics, three of the four *Within* statistics dimension and three of the *Between* statistics dimension reject the null hypothesis of no cointegration. Therefore, in accordance with the economic theory, the standard of living is improving gradually as financial systems are developed in the ECCAS area.

3.4.3. Estimation of the parameters

The method used to estimate our dynamic panel is the system generalized method of moments (System GMM). Indeed, the first difference estimator does not fit with small samples according to Blundell and Bond (1998). In order to overcome this shortcoming, the authors propose an estimator from the system GMM which is associated with two types of tests: (i) The test of Sargan/Hansen for testing the validity of the lagged variables used as instruments; and (ii) The autocorrelation test of Arellano-Bond where the null hypothesis is the absence of autocorrelation of second order of the errors terms in the difference equation.

4. ANALYSIS OF THE RESULTS

The table below shows the one hand there is no second order autocorrelation Arellano and Bond; and secondly, that the instruments are valid. Further, the Wald statistic is very high. This means that the model is generally considered significant.

Table 4.1: Estimation Results by the System GMM (1990-2012)

Dependent: Loggdpc		
Explanatory variables	Coefficients	Z values
Constant	-0826 *	-1.93
Loggdpc L1.	0.858 ***	13.34
Logtop	0.200 **	2.45
Loginv	0.030 *	1.90
Loginfl	0.001 ^{ns}	0.32
Logedu	0.409 *	1.74
Logcrepri	0.0251 ^{ns}	0.64
Logm2	0.014 ^{ns}	0.19
Confl (<i>dummy</i>)	-0073 **	-2.06

Wald: Wald chi 2 (7) = 22863.82; Prob> chi2 = 0.000

Sargan: chi2 (2) = 0.97; Prob> chi2 = 0.617

Arellano-Bond test for AR (1) in first differences: z = -4.64; Pr> z = 0.000

Arellano-Bond test for AR (2) in first differences: z = -1.16; Pr> z = 0.245

Notes: ns = not significant; * = Significant at 10%; ** = Significant at 5%; *** = Significant at 1%;

Source: made with Stata 12 software

From the above results, there is an autocorrelation of the first order of Arellano and Bond test. In fact, the probability of the test AR(1) is 0.0000. However, the probability for the test of absence of autocorrelation of second order (probability test AR (2) equals to 0.245) is greater than 5%. We therefore accept the null hypothesis of the absence of autocorrelation of second order of the error terms in the difference equation of our model.

Also, the probability associated with the Sargan test (probability of test equals to 0.617) is greater than 5%. This test demonstrates that our instruments are valid. So, we can analyze our results

Firstly, financial development indicators (M2 and Credit to private sector) positively influence the level of GDP per capita in the ECCAS. Indeed, the expected signs of the coefficients associated with these variables, in our study, are correct as predicted by the economic theory. However, this positive influence is not significant. We can explain this situation by the fact that financial systems in that region are poorly diversified. The population is very attached to the cash and it does not yet understand the role played by banks in the economy.

One could already predict the non-significance of those financial development indicators for the ECCAS zone. In fact, authors like Berthélemy and Varoudakis (1998) already emphasized that financial development significantly affects economic growth from a certain threshold. According to them, countries should have a M2 to GDP ratio greater than or equal to 36.5%. However, countries in the ECCAS zone have not yet reached that threshold.

Secondly, we observe from the results that the sign of the coefficient associated with inflation is positive. Also, that coefficient is not significant. Indeed, in ECCAS, there is no common monetary policy. The role of the various central banks is to control the level of inflation for macroeconomic stabilization purposes. This result may reflect national disparities in the fighting against inflation in that area. Therefore, in the ECCAS, an increase in the quantity of money in circulation will not result in an increase in inflation. It will not exert adverse effects on the real economy. One can therefore say that certain economies of ECCAS (especially those from the Economic and Monetary Community of Central Africa) have not yet reached the required level of inflation which can boost their economic growth.

Thirdly, the coefficients associated with the trade openness, the level of investment and the human capital are significant and have the expected signs in consistency with the economic theory. So, they are sources of economic growth in the area. In fact, an increase of 10% of the trade openness indicator will result in an increase of 2% in GDP per capita. An increase of 10% in the level of investment, will lead to a growth of the GDP per capita of about 0.3%. An increase of the number of years in the high school of about 10% will lead to a growth rate of 4% in the economy.

Regarding the dummy variable representing conflicts (CONFL), we can see that the presence of conflicts in the zone reduces the level of GDP per capita. The sign and the significance of the coefficient associated to this variable are in line with the predictions of the economic literature. We therefore join the conclusions of the joint report made by RAIAL, Oxfam, and Saferworld (2007) on the harmful effects of conflicts on the standards of living in that region.

Overall, we can see that financial development is not a significant determinant of economic growth in the ECCAS zone. Our result is in line with that of Avom and Carmignani (2010). Also, in the ECCAS zone, the sources of growth are found in: trade openness (international trade), the level of investment (private sector), human capital (education) and conflict resolution (good governance).

However, our results could present some deficiencies. Firstly, it would be interesting to analyze the effects of financial development on the level of GDP per capita in each country of the area. This will help to see national differences and have a deeper comprehension of the phenomenon. Secondly, the use of other indicators could have improved our results. They include: the size of government, the legal origin, the institutional quality, etc.

5. CONCLUSION

The purpose of our analysis was to empirically verify the relationship between financial development and economic growth for ECCAS countries. We therefore used of a dynamic panel modelling for the period 1990-2012. Our study showed more results. Firstly, there is a long-term equilibrium relationship between the banking system and the economic growth in that zone. In fact, the standard of living is improving gradually as financial systems are developed in that zone. Secondly, although having a positive influence on economic growth, financial development does not significantly impact economic growth in that area. We then think about the possible existence of threshold effects in that relationship for ECCAS countries. Thirdly, political instability significantly reduces the standard of living in the zone.

Therefore, the development and the implementation of multisector policies by the ECCAS Member States is a necessity. They can consist in the political stabilization through the strengthening of conflict resolution mechanisms (like the COPAX). There is also a necessity to promote and develop alternative sources of financing such as the stock market, the private equity, the microfinance, etc. It will help to reduce the cost of and access to banking services. Finally, the cooperation and the coordination of monetary policies amongst the different central banks could help solving asymmetric information problems. This will encourage the creation of an optimal currency zone.

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APPENDICES:

Descriptive statistics

<i>Summary Statistics</i>					
Variables	Obs	Mean	Std. Dev.	Min	Max
GDPC	207	1.673265	9.587995	-27.14595	65.77616
M2	207	16.58269	6.197077	4.530003	40.10153
CREPRI	207	8.135888	5.812951	0.7240987	32.46814
TOP	207	83.70943	47.77666	20.43712	275.2324
INV	207	21.91137	16.59465	1.433746	113.5779
EDU	207	6.792271	0.4066655	6	7
CONFL	207	0.2560386	0.4375015	0	1
INFL	207	100.9265	189.6617	7.05e-10	2378.378

Source: made with the Stata12 software

Table 3.2: Results of the unit root test (Im, Pesaran and Shin, 2003)

Variables	W-stat (IPS) ^{at}	P-value	Order of integration
INFL	-3.64989	0.0001	I (0) ^b
GDPC	-5.63726	0.0000	I (1) ^c
M2	-8.64293	0.0000	I (1)
INV	-2.20919	0.0136	I (0)
CREPRI	-8.98179	0.0000	I (1)
EDU	-3.89703	0.0000	I (1)
TOP	-4.33441	0.0000	I (0)

Source: made using data from the WDI, 2013 and Eviews7 software;

Notes: (a) means that it is the statistic unit root test of Im, Pesaran and Shin (2003); (b) means that the variable is integrated of order zero; (c) means that the variable is integrated of order one.

Table 2.3: Results of the test of Pedroni (1999)

Tests	Statistics	P-values
Panel ν^W	-0.819298	0.7937
Rho Panel W	2.631033	0.0043
Panel PP W	-2.404575	0.0081
ADF Panel W	-2.463703	0.0069
Rho Group B	-6.684331	0.0000
PP Group B	-2.952158	0.0016
ADF Group B	-3.688027	0.0001

Source: calculations made via the Eviews7 software .